Calvin Lynch

Graduate Software Test Engineer | BSc (Hons) Software Development
083-097-4469 | calvinlynch181@outlook.com | linkedin.com/in/calvin-l-9620b4321 | cv-calvin.vercel.app

EDUCATION

BSc (Hons) Software Development — First Class Honours *2021 – 2025*
Technological University of the Shannon Midlands Midwest (TUS), Limerick | QQI Level 8 | GPA: 73.66%

CERTIFICATIONS & AWARDS

ISTQB Certified Tester — Foundation Level	ISTQB	2026
QQI Level 8 — Software Development – First Class Honours	TUS	2025
Junior Penetration Tester	TryHackMe	2025
5-Year Service Award	Absolute Hotel Limerick	2023
QQI Level 5 — Software Development (Distinctions in all modules)	LCFE	2021

TECHNICAL SKILLS

Containers & Orch.	Docker, Docker Compose, Kubernetes (K3s), Argo CD, Traefik
CI/CD & Automation	GitHub Actions, Azure DevOps, CI/CD pipeline design
Observability	Prometheus, Grafana, Alertmanager, Node Exporter, PromQL, MELT, Jaeger
Scripting & Backend	Linux, Python, FastAPI, Flask, Java, Spring Boot, PHP, C#, Bash
Databases	InfluxDB, PostgreSQL, CassandraDB, MySQL
Tooling & AI	Git, GitHub, Azure DevOps, Swagger, SpecSync, Claude, AWS Bedrock, OpenRouter
Languages	Python, Java, C#, PHP, HTML, CSS

PROJECTS

Full portfolio: *https://cv-calvin.vercel.app/projects*

AI Vulnerability Monitor — AIPentester

- Developed AIPentester, an AI-assisted penetration testing platform built with Python, FastAPI, PostgreSQL, Docker, and K3s, automating security testing across multiple vulnerability categories.
- Built automated security testing pipelines covering recon, authentication testing, injection attacks, SSRF, IDOR, deserialization flaws, and vulnerability validation workflows.
- Integrated LLM-based agents via OpenRouter to intelligently triage findings, generate attack chains, enrich CVE data with contextual detail, and produce structured pentest reports.
- Designed scalable REST APIs with support for concurrent scan orchestration, scan diffing between runs, per-check result querying, and report export features.
- Deployed using Docker Compose and Traefik on Raspberry Pi/K3s infrastructure; improved scan efficiency through a parallel execution architecture and maintained reliability with 206+ automated tests.

Air Quality Monitoring System — Final Year Project

- Designed and built a portable air quality monitoring system using a Raspberry Pi 5 as the edge computing device, integrating environmental sensors to collect real-time PM2.5, PM10, and CO_2 data.
- Processed all sensor data locally on-device using Python, applying edge computing principles to reduce reliance on cloud processing and minimise latency.
- Exposed collected data via a FastAPI REST API, with time-series data persisted in InfluxDB hosted on AWS, enabling scalable cloud-backed storage and querying.
- Containerised all backend services using Docker for consistent, portable deployments across development and production environments.
- Integrated Prometheus and Grafana for infrastructure-level monitoring and observability of the full deployed stack, including alerting configuration.
- Built a React web application and a Swift iOS mobile app to provide real-time data visualisation, giving users cross-platform access to live and historical air quality readings.

Monitoring & Alerting Stack

- Designed and self-hosted a full observability stack on a Raspberry Pi using Docker, covering metrics collection, dashboard visualisation, and automated alerting end-to-end.
- Configured Prometheus with custom scrape jobs targeting system services; wrote PromQL queries to evaluate real-time performance metrics and define alert thresholds.
- Deployed Node Exporter to expose host-level system metrics (CPU, memory, disk, network) for collection by Prometheus.
- Built Grafana dashboards to visualise live and historical metrics, providing clear system health overviews across all monitored services.
- Configured Alertmanager with custom alerting rules and webhook integrations to route notifications on service failure; validated the full alerting pipeline through deliberately simulated downtime scenarios.

Task Management System *2024*

- Designed and developed a secure Task Management System using Java Spring Boot, exposing RESTful APIs for full CRUD functionality across task resources.
- Integrated Spring Security with JWT-based authentication, OAuth2 support (Google login), and OTP-based multi-factor authentication to provide layered access control.
- Configured HTTPS using self-signed TLS certificates to secure all API communication, and documented all endpoints using Swagger for developer usability.
- Implemented a comprehensive automated test suite using JUnit and Mockito, integrated into a GitHub Actions CI/CD pipeline with JaCoCo for code coverage tracking and reporting.

WORK EXPERIENCE

Graduate Software Test Engineer — Release & Regression Team *July 2025 – Present*

Kneat Solutions

- Co-developed as part of a team an AI-assisted PR security analysis platform using Claude, OWASP Top 10, and CVE-based vulnerability detection — awarded 1st place out of 16 teams in an internal AI Engineering competition. Enabled faster PR reviews and automated senior Security Manager escalation based on vulnerability severity.
- Co-developed an automated test reporting platform using Python, Poetry, Azure DevOps REST APIs, and CI/CD pipelines. Integrated Claude-based AI agents via AWS Bedrock to analyse test plans, validate compliance, generate customer-ready reports, and support requirements traceability using SpecSync.
- Execute regression, exploratory, and PR testing; logging and validating defects using Azure DevOps and Git across multiple application modules.
- Achieved ISTQB Certified Tester Foundation Level certification and completed internal Kneat End User and Power User Level 1 training.

Software Engineer Intern — Platform Operations Team *Jan 2024 – Aug 2024*

Progress Software Corporation (8 months)

- Embedded within a Platform Operations team gaining direct experience with DevOps practices, CI/CD pipelines, automation, and Agile delivery (SAFe framework).
- Built a cloud cost analysis dashboard integrating data from Azure CLI, CassandraDB, and Prometheus, visualised in Grafana for internal use — end-to-end ownership from data pipeline to dashboard.
- Implemented MELT observability (Metrics, Events, Logs, Traces) using Prometheus for metric scraping and alerting across platform services.
- Deployed a Jaeger instance for distributed tracing and service dependency analysis.
- Daily stack: Python, Docker, Kubernetes (K3s), GitHub Actions, FastAPI, Flask, CassandraDB, InfluxDB, Grafana, Argo CD.
- Participated in sprint reviews, PI planning, code reviews, and retrospectives.

Senior Receptionist & Night Duty Manager *2018 – 2024*

Absolute Hotel, Limerick

- Progressed from Part Time Receptionist (2017 - 2018) to Night Duty Manager (2018–2019) to Senior Receptionist over a 6 - year tenure. Awarded 5 -Year Service Award (2023).